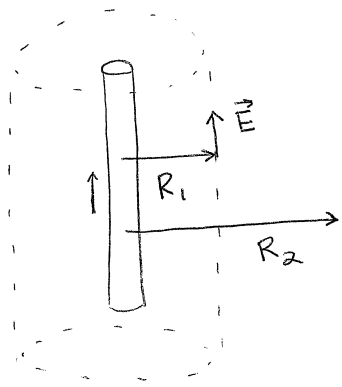
#3



$\vec{E}(R_1) = \hat{z} E_1 \cos(\omega t)$

$\vec{E}(R_2) = ?$

1D: $\varphi(t,x) = f(t - \frac{x}{c}) + g(t - \frac{x}{c})$

3D: $\varphi(t,r) = \dfrac{f(t-\frac{r}{c})}{r} + \dfrac{g(t+\frac{r}{c})}{r}$

In this problem, we expect the average radiated power to be constant.

$$\langle P \rangle = \int \langle \vec{S} \rangle \cdot d\vec{a} = \langle S \rangle A \propto E^2 A = E^2 (2\pi r L) = \text{constant}$$

From this we might guess $E \sim \dfrac{1}{\sqrt{r}}$

We are free to let $V = 0$ so that $\vec{E} = -\dfrac{\partial \vec{A}}{\partial t}$

Try $\vec{A} = \dfrac{X(t,\rho)}{\sqrt{\rho}} \hat{z}$

Our trial solution should satisfy the differential equation $\dfrac{1}{c^2}\dfrac{\partial^2 \vec{A}}{\partial t^2} - \nabla^2 \vec{A} = 0$

$$\nabla^2 \vec{A} = \frac{1}{\rho}\left(\frac{\partial}{\partial \rho}\left(\rho \frac{\partial \vec{A}}{\partial \rho}\right)\right)$$

$$\frac{\partial A_z}{\partial \rho} = \frac{\partial}{\partial \rho}\left(\rho^{-1/2} X\right) = -\frac{1}{2}\rho^{-3/2} X + \rho^{-1/2}\frac{\partial X}{\partial \rho}$$

$$\frac{\partial}{\partial \rho}\left(\rho \frac{\partial A_z}{\partial \rho}\right) = \frac{\partial}{\partial \rho}\left(-\frac{1}{2}\rho^{-1/2} X + \rho^{1/2}\frac{\partial X}{\partial \rho}\right)$$

$$= \frac{1}{4}\rho^{-3/2}X - \cancel{\frac{1}{2}\rho^{-1/2}\frac{\partial X}{\partial \rho}} + \cancel{\frac{1}{2}\rho^{-1/2}\frac{\partial X}{\partial \rho}} + \rho^{1/2}\frac{\partial^2 X}{\partial \rho^2}$$

$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial A_z}{\partial \rho}\right) = \frac{X}{4\rho^{5/2}} + \frac{1}{\rho^{1/2}}\frac{\partial^2 X}{\partial \rho^2}$$

$$\frac{1}{c^2}\frac{\partial^2 A_z}{\partial t^2} - \nabla^2 A_z = \frac{1}{c^2}\frac{1}{\sqrt{\rho}}\frac{\partial^2 X}{\partial t^2} - \frac{1}{\sqrt{\rho}}\frac{\partial^2 X}{\partial \rho^2} - \frac{1}{4\rho^{5/2}}X = 0$$

$$\Rightarrow \frac{1}{c^2}\frac{\partial^2 X}{\partial t^2} - \frac{\partial^2 X}{\partial \rho^2} - \frac{1}{4\rho^2}X = 0$$

Provided that ρ is large, so that the last term is neglibable, our trial solution is a general solution.

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} = -\frac{1}{\sqrt{\rho}}\frac{\partial X}{\partial t}\hat{z}$$

At $\rho = R_1$

$$-\frac{1}{\sqrt{R_1}}\frac{\partial X}{\partial t} = E_1\cos(\omega t) \quad \Rightarrow \quad X = -\frac{E_1\sqrt{R_1}}{\omega}\sin\omega t$$

$$\vec{A} = -\frac{E_1}{\omega}\sqrt{\frac{R_1}{\rho}}\sin\omega t\,\hat{z}$$

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} = E_1\sqrt{\frac{R_1}{\rho}}\cos\omega t\,\hat{z}$$

Remembering to use retarded time and introducing a phase so that our boundary condition is still satisfied:

$$\vec{E}(\rho, t) = E_1\sqrt{\frac{R_1}{\rho}}\cos\left(\omega(t-\rho/c) + \frac{\omega R_1}{c}\right), \quad \vec{E}(\rho = R_2, t) = E_1\sqrt{\frac{R_1}{R_2}}\cos\left(\omega\left(t - \frac{R_2}{c}\right) + \frac{\omega R_1}{c}\right)$$

In order to be able to neglect $\frac{1}{4\rho^2}X$ in our differential equation,

$$\left|\frac{X}{\rho^2}\right| \ll \left|\frac{\partial^2 X}{\partial \rho^2}\right| \quad , \quad \frac{\partial^2 X}{\partial \rho^2} \sim k^2 X$$

$$\Longrightarrow \quad \frac{X}{\rho^2} \ll k^2 X$$

$$\Longrightarrow \quad \left(\frac{1}{k\rho}\right)^2 \ll 1 \quad \text{or since} \quad k \sim \frac{1}{\lambda}, \quad \lambda \ll \rho$$